

Second-round verification of the first agent’s work on

“Gravitational-wave perturbations of a spherical thin shell:
junction conditions, scattering coefficients and quasinormal modes”

Verification report (second round)

25 September 2026

Abstract

This report documents every check made on the work contained in the folder **First round** (the derivation document **derivation/derivation.pdf**, the Python code in **code/**, the results in **results/**, the tables and the twelve figures, and the **README.md**). For each claim we state how it was verified – by re-running the first agent’s own code, by an independent re-derivation, by an independent code written for this purpose, or by reading the cited literature – and the verdict. The first round is reproducible bit for bit and its main results for the odd sector and for the static fluid shell (model C) are confirmed independently, including a new thick-shell derivation of the odd junction condition and a new Newtonian derivation of the shell’s matter modes. We find one substantive physical problem: the even-parity results of the adiabatically frozen domain wall (model A) are *not* a well-defined $O(\omega_{\text{dyn}}/\omega)$ approximation, but depend at $O(1)$ on which of the Israel equations are imposed. We also find a number of overstated accuracy claims, catalogue omissions and mislabels, a wrong author name, and a misprint in the reference paper PSP26. No file of the first round was modified (verified by SHA-256 hashes).

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1 Scope, material and protocol

Material examined. The task statement PROMPT.md; the prior notes Starting point/interior_ondas_cascara.pdf and burbuja_matching_israel.pdf; the seven papers of Bibliography/; and everything in First round/: derivation.tex/.pdf (958 lines), ten L^AT_EX tables, the code (main.py, make_figures.py, make_tables.py, polish_catalogue.py, package shellgw/ with 14 modules, symbolic/ with 5 scripts; the two auto-generated even-parity junction modules are 1.0 and 1.6 MB of sympy output), the result files (validation.json, scattering.json, qnm_catalogue.json, qnm_tracks.json, qnm_table.csv, 22 log files) and 12 figures (PDF+PNG). All twelve figures were inspected visually.

Non-interference. Before any action the SHA-256 hash of every file of First round was stored in Second round/first_round_sha256_before.txt (102 files) and compared at the end (Sec. 13). The first round's code writes figures and results through relative paths (shellgw/plots.py, studies.py); it was therefore *never* executed in place. All executions used copies: Second round/rerun, rerun_validate, rerun_evenC (execution) and fr_code_pristine (read-only import of the generated junction matrices by the independent codes).

Independent material written in the second round (folder Second round/checks/; each script writes an out_*.txt/json file):

script	content
c01_background.py	symbolic re-derivation of the background Israel equations, I&S (3.7)–(3.9), static shell, dust, domain wall, energy/causality bounds, Table 1 numbers.
c02_blackhole.py	own Leaver continued fraction (mpmath, 30 digits); own BH reflectivity (LSODA in r_*); symbolic check of the asymptotic recursion; explicit Iyer–Will 3rd-order WKB formula.
c03_axial_thick_shell_derive.py	from-scratch linearisation of the Einstein equations for axial perturbations of a static <i>anisotropic</i> fluid; master potential.
c03b_axial_thick_shell_numeric.py	smooth thick shell ($p_r = 0$) $\rightarrow$ thin shell limit of the odd junction.
ind_solver.py	independent thin-shell solver (mpmath Bessel functions, LSODA, Zerilli asymptotics without the Chandrasekhar map, different complex ray, secant root finder).
c04_compare_scattering_qnm.py	$\mathcal{R}$, $\mathcal{T}$ and QNMs recomputed and compared with the first round.
c05_newtonian_shell.py	from-scratch Newtonian theory of the non-radial oscillations of a self-gravitating fluid shell.
c06*_.py	properties of the even-parity transfer matrices; model-A prescription dependence.
c07_gamma_dependence.py	Γ dependence of the matter modes (vs PSP26 Figs. 1–2).
c08_limits.py	trapped modes, $R \rightarrow 2M$, low frequency, closed system, NaN point.
c09_poles_and_missing_modes.py	poles of T_{even} ; matter modes missing from the catalogue.
integrity_check.sh	SHA-256 comparison of the first round before/after.
make_round2_figures.py	the five figures of this report.

Verdict labels. **CONFIRMED**: verified to the stated accuracy by at least one independent means; **PARTLY**: correct in substance but imprecise, incomplete or only partially checkable; **OVER-STATED**: numerically true only in a weaker form; **INCORRECT**: wrong; **NOT VERIFIABLE**: could not be checked with the available sources.

2 Summary of findings

2.1 Main conclusions

1. **Reproducibility** is complete: re-running the first round’s scripts reproduces the symbolic logs, the whole `validation.json` (487 numbers identical to the last bit), all ten L^AT_EX tables (identical files) and all twelve figures (pixel-identical).
2. **Background and choice of model**: all formulas are correct (Sec. 4). The rejection of “Option B” (static dust) is correct, but the justification “the formula $M = 4\pi\sigma R^2\sqrt{f}$ does not follow from the junction conditions” is imprecise: that formula *is* the static pressure-balance condition with $p = 0$; it is incompatible with the other junction condition.
3. **Odd-parity junction** $\Psi_+ = \Psi_-$, $\sqrt{f_R}\Psi'_+ - \Psi'_- = (1 - \sqrt{f_R})\Psi/R$ is **CONFIRMED** by three independent routes: (i) a line-by-line re-derivation; (ii) a new derivation of the axial master potential of a smooth anisotropic fluid, $V = e^{2\Phi}[\ell(\ell+1)/r^2 - 6m/r^3 + 4\pi(\rho - p_r)]$, whose thin-shell limit gives exactly $\Delta_{\text{odd}} = 4\pi\sigma\sqrt{f_R}$; (iii) a numerical thick-shell→thin-shell limit, in which the first round’s junction converges linearly in the thickness while three wrong alternatives do not (Fig. 2).
4. **Even-parity junction, model C**: **CONFIRMED**. $\det T\sqrt{f_R} = 1$ to 10^{-12} , T real and flux-conserving for real ω , transparent limit linear in M . The matter modes agree with PSP26 Figs. 1–2 (read by us) at all compactnesses and also as functions of Γ (a comparison the first round did not make), and tend to the Newtonian limit that we derived independently (Fig. 3).
5. **New finding on the reference paper**: PSP26 Eq. (7.6) contains a misprint (the constant term must be $-(\ell-1)\ell(\ell+1)[(\ell+2)(2\ell-3)\Gamma - 4(\ell-2)]/[16(2\ell+1)]$); their Eq. (7.8), used by the first round, is correct.
6. **Model A (frozen domain wall), even parity**: **INCORRECT** as an approximation with controlled error. Keeping all four metric-continuity equations, the choice of which two of the four Israel equations are imposed changes $\mathcal{R}(\omega)$ by orders of magnitude even at $\omega M = 30$ (Fig. 4), and the lowest QNMs by 20–30%. With the first round’s choice the energy error $|\mathcal{R} + \mathcal{T} - 1|$ is *larger than $\mathcal{R}$ itself* at all high frequencies, so its $\mathcal{R}$ has no significant digit. One alternative choice ($\tau\tau$ + trace-free) conserves energy exactly and reproduces the universal δ -barrier law of the odd sector and of model C. The statement “the frozen domain wall reflects much more strongly” is therefore an artifact of the prescription. Moreover the wall collapses on the light-crossing time ($\omega_{\text{dyn}} \simeq \sqrt{2}/R$), not on $(R^3/M)^{1/2}$ as stated in the Discussion, so the wall’s wave QNMs ($|\omega| \approx 2\text{--}5\omega_{\text{dyn}}$) are not in the adiabatic regime (Fig. 1).
7. **Numerics**: $\mathcal{R}, \mathcal{T}$ reproduced by an independent solver to 10^{-11} (and $\mathcal{T}$ to 10^{-8}); every QNM tested reproduced to $10^{-10}\text{--}10^{-13}$ (damped imaginary matter modes to $10^{-5}\text{--}10^{-7}$). The accuracy claims “QNMs agree to $\lesssim 10^{-8}$ ” (README) and “ $10^{-8}\text{--}10^{-13}$ ” (derivation) are **OVERSTATED**: 28 of 397 catalogue modes differ by more than 10^{-8} between the two methods, the worst by 6.4×10^{-5} (still within the required 10^{-4}).
8. **Catalogue**: incomplete for compact shells – the stable matter pair is missing for $\ell = 2$ at $R = 2.2, 2.5, 3M$ (it exists: e.g. $0.197121 - 0.000310i$ at $R = 3M$) and the unstable $\ell = 4$ mode is missing at $R = 2.2M$ ($0.43726i$); two modes are mislabelled.
9. **Literature**: all citations checked exist and are used correctly, except the author of the eikonal instability paper (Yang, Bonga and Z. Pan, PRL 130, 011402 (2023), not “U.-L. Pen”).

2.2 Claim-by-claim table

ID	claim of the first round	how checked	verdict
<i>Reproducibility</i>			
R1	symbolic scripts reproduce the logs	re-run in copy	CONFIRMED

ID	claim of the first round	how checked	verdict
R2	<code>validation.json</code> , tables, figures follow from the code	re-run, diff, pixel compare	CONFIRMED
R3	generated junction modules follow from <code>derive_junction.py</code>	re-run (Sec. 3)	CONFIRMED
<i>Background (derivation Sec. 1)</i>			
B1	$\sigma = (1 - \sqrt{f_R})/4\pi R$, $p = (1 - M/R - \sqrt{f_R})/(8\pi R\sqrt{f_R})$ (P04 3.80–3.81)	sympy (c01) + P04 text	CONFIRMED
B2	Israel $\Leftrightarrow$ I&S (3.7a,b); mass formula (3.8), (3.9)	sympy (c01); IS84 page image	CONFIRMED
B3	static shell needs $\tau/\sigma = -\kappa$, $\kappa = (1 - \sqrt{f_R})/4\sqrt{f_R}$	sympy	CONFIRMED
B4	DEC $\Leftrightarrow R \geq 25M/12$; $\Gamma = 2$ causality $\Leftrightarrow$ same bound	sympy	CONFIRMED
B5	no static dust shell	sympy	CONFIRMED
B6	“ $M = 4\pi\sigma R^2\sqrt{f_R}$ does not follow from the junction conditions”	sympy	PARTLY
B7	dust turning point: same geometry, $\ddot{R} = -M/[R^2(1 + \sqrt{f_R})]$	sympy	CONFIRMED
B8	wall: $\ddot{R} = -M/[R^2(1 + \sqrt{f_R})] - 2\sqrt{f_R}/R = -(1 + 3\sqrt{f_R})/(2R)$	sympy; also first-round log	CONFIRMED
B9	Table 1 (background.tex) numbers	recomputed	CONFIRMED
B10	I&S quote “ $\ddot{R}$ always negative if $\tau \geq 0$ ”	IS84 p. 714	CONFIRMED
B11	interior frequency $\nu = \omega/\sqrt{f_R}$ (correction of prompt and prior notes)	P04 (3.75); PSP26	CONFIRMED
B12	“every thin shell of this type evolves on $(R^3/M)^{1/2}$ ”; “growth time comparable to wall collapse”	Table 1, c01	INCORRECT
<i>Perturbation equations (Sec. 2–3)</i>			
P1	V_Z (Zerilli Eq. 5) = $f \times \text{MP05}$ (4.26)	re-run (residual 0)	CONFIRMED
P2	odd reconstruction $h_t = \frac{f}{2}(r\Psi)'$, $h_r = -i\omega r\Psi/2f$	re-run (5 residuals 0); MP05 (5.18)	CONFIRMED
P3	even reconstruction Eqs. (2.8)–(2.10), $Q(r)$	re-run + sympy comparison with the printed formulas	CONFIRMED
P4	first-order system agrees with Zerilli (1970)	compared with Z70 p. 737	CONFIRMED
P5	Chandrasekhar map, amplitude factor $\mu(\mu + 2) + 12iM\omega$	re-run; analytic	CONFIRMED
P6	Riccati–Bessel/Hankel functions, Wronskian = 1	mpmath (ind_solver)	CONFIRMED
<i>Odd junction (Sec. 4.2)</i>			
O1	δK_{tA} , δK_{AB} (Eq. 4.4)	hand re-derivation	CONFIRMED
O2	AB equation $\Rightarrow \Psi_+ = \Psi_-$; $\tau A \Rightarrow U = 0$; $[K]^{(0)}$ and p cancel	hand	CONFIRMED
O3	final junction (4.8), $\Delta_{\text{odd}} = \sqrt{f_R}(1 - \sqrt{f_R})/R$, $\det \mathbb{T} = f^{-1/2}$	thick-shell theory + numerics (c03)	CONFIRMED
O4	agreement with PSP26 (5.23) and Pani et al. (3.29), (A35)	read both papers	CONFIRMED
O5	model-A odd system consistent (rank 3)	first-round log	CONFIRMED
<i>Even junction (Sec. 4.3)</i>			
E1	$\det \mathbb{T}^C = f_R^{-1/2}$ (10^{-12} double, 10^{-15} mp)	300 random samples (c06)	CONFIRMED
E2	flux conservation, $\mathcal{R} + \mathcal{T} = 1$	$\mathbb{T}$ real, flux 10^{-13} (c06)	CONFIRMED
E3	transparent limit 4×10^{-9} at $M = 10^{-8}$	c06	CONFIRMED
E4	double precision error 10^{-6} at $R = 2.2M$, breakdown $R \lesssim 2.05M$	c06: 1.3×10^{-7} ; 5×10^{-3} at 2.05	PARTLY
E5	$\mathbb{T}$ has poles at the matter-mode frequencies	zeros of $\det A$ (c09)	PARTLY
E6	matter modes reproduce PSP26 Figs. 1–2 and PN limit Eq. (7.8)	PSP26 read; own solver; own Newtonian theory	CONFIRMED
E7	wave modes agree with PSP26 Figs. 3–4 to ± 0.02	PSP26 read	CONFIRMED
E8	even-C merges with odd at high frequency (δ -barrier law)	own solver (c06c)	CONFIRMED

ID	claim of the first round	how checked	verdict
<i>Model A (Sec. 4.3, 6)</i>			
A1	rank $A = 6$, rank $[A B] = 8$	first-round log	CONFIRMED
A2	constraints violated $\propto \omega^{-1}$, $ \mathcal{R} + \mathcal{T} - 1 \propto \omega^{-2}$	c06 (exponent -2.00)	CONFIRMED
A3	$\ddot{T}_0 \propto \ddot{R}^2 \neq 0$ at the turning point	analytic	CONFIRMED
A4	even-A results “ $O(\omega_{\text{dyn}}/\omega)$ -accurate”	six prescriptions (c06d, c06e)	INCORRECT
A5	“frozen domain wall reflects much more strongly”	same	INCORRECT
<i>Numerics (Sec. 5–6)</i>			
N1	continued fraction (Pani B4 with $\omega \rightarrow -\omega$), convergence iff $R_2 \geq 4M$	re-run; Pani App. B	CONFIRMED
N2	asymptotic recursion Eq. (5.5)	sympy (c02)	CONFIRMED
N3	M1 vs M2 on $\mathcal{R}, \mathcal{T}$: $\lesssim 10^{-10}$	table: 6×10^{-10} ; own solver	PARTLY
N4	M1 vs M2 on QNMs: 10^{-8} – 10^{-13} / README $\lesssim 10^{-8}$	catalogue: max 6.4×10^{-5}	OVERSTATED
N5	QNM values	own solver, 10^{-10} – 10^{-13}	CONFIRMED
N6	catalogue/tables list the lowest modes	own searches	PARTLY
N7	Boyanov et al. estimator implemented, agrees within $O(1/r)$	table (2 – 4×10^{-6})	CONFIRMED
N8	NaN at $\ell = 4$, $\omega M = 0.0057$	scattering.json	CONFIRMED
<i>Schwarzschild validation and limits</i>			
S1	$\ell = 2, n = 0$: $0.37367168 - 0.08896232i$	own Leaver (30 digits)	CONFIRMED
S2	$n \leq 3$, $\ell = 2, 3, 4$ agree with BCS09 to 6 digits	own Leaver vs 6-digit table	CONFIRMED
S3	WKB orders 1, 3, 6; IW Table I values	explicit IW formula (order 3)	CONFIRMED
			(3) / NOT VERIFIABLE
S4	quoted WKB errors 7×10^{-2} , 1.5×10^{-3} , 2.3×10^{-4}	recomputed: these are <i>relative</i> errors	PARTLY
S5	BH reflectivity 1.55×10^{-2} , 5.3×10^{-9} ; Vishveshwara Fig. 2	own integrator; V70 scan	CONFIRMED
L1	low frequency: $1 - \mathcal{R} < 10^{-11}$	own solver: 10^{-15}	CONFIRMED
L2	high frequency: $\mathcal{R} \simeq \Delta^2/4\omega^2$ (odd)	own solver: ratio $\rightarrow 1$	CONFIRMED
L3	$\mathcal{R} \rightarrow \mathcal{R}_{\text{BH}}$ as $R \rightarrow 2M$	own solver down to $R = 2.0002M$	PARTLY
L4	closed system $ S = 1$	own solver: 10^{-16}	CONFIRMED
L5	trapped modes; WKB Re to 1–6%, Im within factor $\lesssim 2$	own solver (10^{-12}); 0.6–6.3%, up to 2.03	PARTLY
<i>Literature and task compliance</i>			
C1	PSP26 exists, same system, equation numbers and content	PDF read	CONFIRMED
C2	Boyanov et al., PRL 135, 151402 (2025)	INSPIRE	CONFIRMED
C3	other citations (Tab. 3)	INSPIRE/ADS/local PDFs	CONFIRMED
C4	“H. Yang, B. Bonga and U.-L. Pen”	APS/INSPIRE	except one INCORRECT
C5	every function has a docstring citing an equation	AST audit	PARTLY
C6	required plots, CSV columns, tolerances	inspection	CONFIRMED

3 Reproducibility

Symbolic scripts. In the copy Second round/rerun we executed `symbolic/vacuum_relations.py` (33s) and `symbolic/exterior_series.py` (1.5s). The outputs (`rerun/rerun_log_vacuum.txt`, `rerun_log_exterior_series.txt`) coincide with the first-round logs `log_vacuum_relations.txt` and `log_exterior_series.txt`: the Zerilli/MP05 potential identity, the five odd vacuum residuals, the CPM reconstruction, the even first-order system, the algebraic identity, the reconstruction formulas and the “max residual of all 6 vacuum Einstein equations ... = 0” are all reproduced, as is the c, d, e polynomial check and the Chandrasekhar-map residual 0. `derive_junction.py` was re-run for all four

cases (`rerun/`) and separately for the even model-C case (`rerun_evenC/`); see the note at the end of this section for its outcome.

Numerical stage. `main.py validate` was executed in `rerun_validate` (38 s). The new `validation.json` was compared recursively with the original: all 487 numbers are identical to the last bit. `make_tables.py` regenerates the ten tables of the derivation: all ten files are byte-identical. `make_figures.py` regenerates the twelve figures from the stored results: all PNGs are pixel-identical. The long stages (`scatter`, 546 s, and `qnm`, 3283 s, on 10 cores) were not re-run as a whole; instead their outputs were *recomputed with independent code* (Secs. 9), which is a stronger test.

Junction-module regeneration. `derive_junction.py` was re-run in fresh copies. For the even sector (`even:C`, 32 s; `even:A`, 43 s) the log lines of the first round are reproduced verbatim (background residual 0, $\ddot{R}$ for the wall, ranks 8/8 and 6/8, residual 4.62×10^{-15} and 7.12×10^{-3} , the test matrices T), the pickled symbolic matrices A, B (and $\ddot{R}$) are *structurally identical* to the first round's, and the exported Python modules are identical apart from the UTF-8 byte-order mark and the final line ending. For the odd sector the full script did not finish within 35 minutes: after exporting the matrices it attempts an optional symbolic `sympy.solve/simplify` of the system (whose output is only printed; the first-round log also stops before that line). Calling the script's `derive()` function directly, the odd A, B of models C and A are obtained in 1 s; they differ from the stored ones only in the printed form of the expressions and agree numerically to 1.5×10^{-13} (the resulting T to 3×10^{-15}) at random complex parameters. Hence all four generated junction systems are reproducible (R3 **CONFIRMED**).

4 Background spacetime, collapse problem and choice of model

Method. In `c01_background.py` the jumps of the mixed extrinsic curvature, $K^\theta_\theta|_\pm = \{\beta, \alpha\}/R$, $K^\tau_\tau|_+ = (\ddot{R} + M/R^2)/\beta$, $K^\tau_\tau|_- = \ddot{R}/\alpha$ [IS84 (3.5)–(3.6), whose page we rendered and read], were inserted into $[K^a_b] - \delta^a_b[K] = -8\pi S^a_b$ with $S^a_b = \text{diag}(-\sigma, p, p)$ and solved for σ, p .

Results.

- I&S (3.7a) and (3.7b) (with $\tau = -p$) are satisfied identically by these σ, p (numerical residuals 5×10^{-17} and 0 at a random point with $\dot{R} \neq 0$); the mass formula (3.8) holds to 10^{-16} ; at $\dot{R} = 0$, $4\pi\sigma R = 1 - \sqrt{f_R}$ and (3.9) holds exactly (B2 **CONFIRMED**).
- Static shell: $\sigma = (1 - s)/4\pi R$, $p = (1 - s)^2/(16\pi R s)$ with $s = \sqrt{f_R}$; this equals Poisson's (3.81) (residual 0) and $\kappa = p/\sigma = (1 - s)/4s = M/[2Rs(1 + s)]$ (B1, B3 **CONFIRMED**).
- $\kappa = 1$ and $v_s^2 = 2\kappa/(1 + \kappa) = 1$ both occur at $s = 1/5$, i.e. $R = 25M/12$ (B4 **CONFIRMED**).
- Dust: $p = 0$ with $\dot{R} = \ddot{R} = 0$ forces $s = 1$ ($M = 0$) (B5 **CONFIRMED**). At a turning point $\ddot{R} = -(1 - s)/2R = -M/[R^2(1 + s)]$ (B7 **CONFIRMED**).
- Domain wall ($p = -\sigma$) at $\dot{R} = 0$: $\ddot{R} = -(1 + 3s)/2R$, equal to $-M/[R^2(1 + s)] - 2s/R$ (B8 **CONFIRMED**). All nine columns of Table 1 (`background.tex`) are reproduced to all printed digits (B9 **CONFIRMED**).

Remark on Option B (B6, **PARTLY).** For a static shell the pressure equation $8\pi p = [K^\tau_\tau] + [K^\theta_\theta]$ with $p = 0$ gives $[K^\tau_\tau] = M/(R^2\sqrt{f_R}) = 4\pi\sigma$, i.e. exactly the task's formula $M = 4\pi\sigma R^2\sqrt{f_R}$. So that formula *does* follow from one junction condition; it is the second one, $4\pi\sigma R = 1 - \sqrt{f_R}$, that is incompatible with it unless $M = 0$. The first round's conclusion is right, its wording is not.

Time scales (B12, **INCORRECT).** The Discussion states that “every thin shell of this type evolves on the timescale $(R^3/M)^{1/2}$ ” and that the model-C growth time $\sim 1.6(R^3/M)^{1/2}$ “is comparable to the collapse time of the domain wall”. From $\ddot{R} = -(1 + 3\sqrt{f_R})/2R$ one has

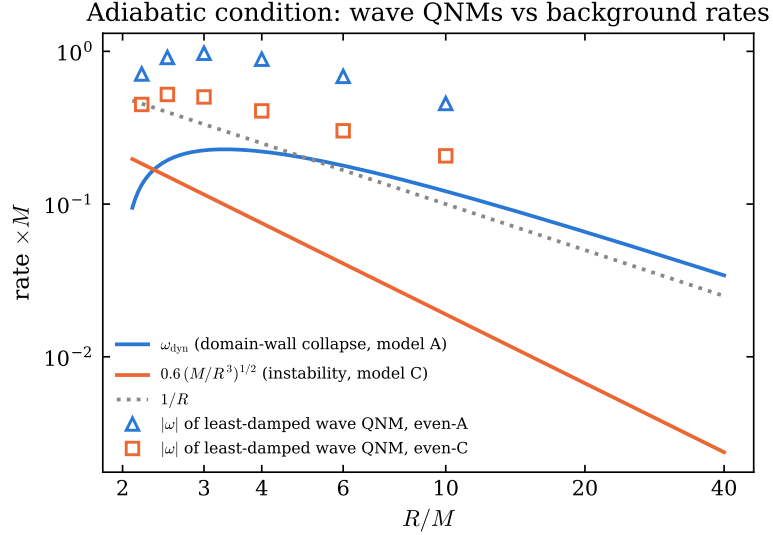


Figure 1: Background rates versus the least-damped wave QNMs ($\ell = 2$). The domain-wall collapse rate ω_{dyn} follows $\sim 1/R$ and is only a factor 2–5 below the model-A wave QNMs, whereas the model-C instability rate $\propto (M/R^3)^{1/2}$ is well separated.

$\omega_{\text{dyn}} = \sqrt{f_R}(|\ddot{R}|/R)^{1/2} \rightarrow \sqrt{2}/R$ for $R \gg M$: a tension-dominated wall collapses on the light-crossing time, independently of M . At $R = 6M$ the wall’s collapse time is $1/\omega_{\text{dyn}} = 5.6M$ while the model-C growth time is $1/0.0401 = 25M$; at $R = 10M$, $8.2M$ vs $57M$. Fig. 1 shows that ω_{dyn} is of the same order as the wave-mode frequencies, so the argument “wave-mode QNMs have $\omega \sim 1/R \gg (M/R^3)^{1/2}$ ” covers model C but not model A.

Blueshift (B11, CONFIRMED). Continuity of h_{ab} at a static shell gives $dT = \sqrt{f_R} dt$ (Poisson’s interior metric (3.75), which we read); PSP26 use $w = F^{-1/2}\omega$ inside. The correction to the task statement and to `interior_ondas_cascara` (which uses $\hat{j}_\ell(\omega r)$) is justified.

5 Perturbation equations and interior solution

The re-run of `vacuum_relations.py` gives zero residuals for all statements of derivation Sec. 2 (P1, P2). We checked that the *printed* reconstruction formulas (2.8)–(2.10), including the polynomial $Q(r)$, are algebraically identical to the expressions printed in the log (difference simplified to 0 with sympy) (P3). The K' equation of the first-round first-order system coincides term by term with Zerilli’s (1970, p. 737) system for $e^{-i\omega t}$ (P4). The Chandrasekhar map is verified by the re-run (residual 0) and for outgoing waves $Z \rightarrow [\mu(\mu+2) + 12iM\omega]e^{i\omega r_*}$ follows from $d\Psi/dr_* = i\omega\Psi$ (P5). Using mpmath Bessel functions and the exact recurrence $(zj_\ell)' = zj_{\ell-1} - \ell j_\ell$ we verified the Wronskian $\hat{j}\hat{y}' - \hat{j}'\hat{y} = 1$ (to 10^{-16} at real and complex argument) and agreement with `scipy.special` to 10^{-15} (P6).

6 Odd-parity junction condition

6.1 Line-by-line re-derivation

We re-derived by hand the chain of derivation Sec. 4.2: (a) the covariant form of $\delta\Gamma^r_{t\phi} = \frac{f}{2}(\partial_t h_r - \partial_r h_t)X_\phi$ (the Γ -terms of the three covariant derivatives cancel pairwise) and $\delta\Gamma^r_{AB} = fh_r X_{AB}$ (using $D_A X_B + D_B X_A = 2X_{AB}$); with $n_r = f^{-1/2}$ unperturbed this gives Eq. (4.4). (b) With the CPM definition $\partial_r h_t - \partial_t h_r = \mu\Psi/2r + 2h_t/r$, the proper-time components (4.5) follow. (c) $\delta S_{AB} = 0$ gives $[\sqrt{f_R}h_r] = 0$, and with $h_r^+ = -i\omega R\Psi_+/2f$, $h_r^- = -i\nu R\Psi_-/2$, $\nu = \omega/\sqrt{f_R}$, one gets $\Psi_+ = \Psi_-$ exactly (for $\omega \neq 0$). (d) In the τA equation the combination $[K]^{(0)}\mathcal{H} - 8\pi p\mathcal{H} = -4\pi\sigma\mathcal{H}$

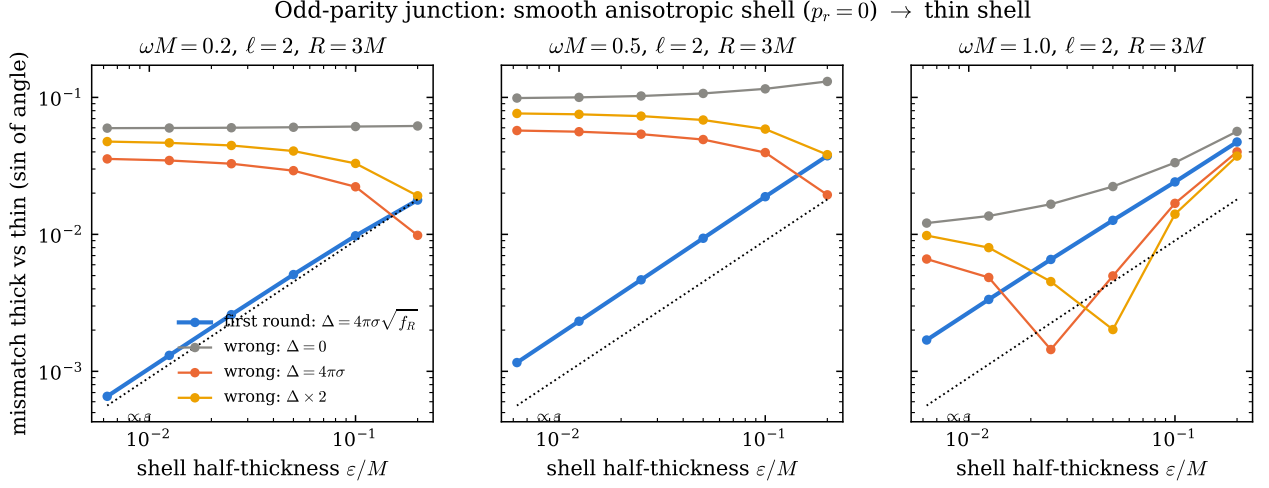


Figure 2: Independent test of the odd junction: smooth anisotropic shells of decreasing thickness converge to the first round's thin-shell junction and to none of the alternatives.

is independent of p (hence of $\ddot{R}$), and $(\sqrt{f_R} - 1)/R = -4\pi\sigma$ cancels it: $U = 0$. (e) Continuity of $\mathcal{H} = \frac{1}{2}(r\Psi_-)' = \frac{\sqrt{f_R}}{2}(r\Psi_+)'$ gives (4.8). All steps are correct (O1, O2 **CONFIRMED**).

6.2 Independent derivation: axial perturbations of an anisotropic fluid

Because (a)–(e) share the first round's framework, we derived the odd junction a second time in a completely different way. In `c03_axial_thick_shell_derive.py` we linearised the Einstein equations (own code) for the static metric $-e^{2\Phi}dt^2 + dr^2/(1 - 2m/r) + r^2d\Omega^2$ with an anisotropic fluid $T_{\mu\nu} = (\rho + p_t)u_\mu u_\nu + p_t g_{\mu\nu} + (p_r - p_t)s_\mu s_\nu$ (s normal to the matter layers) and an axial perturbation (metric h_0, h_1 and fluid velocity U). Eliminating h_0 with the $\theta\phi$ equation, the $r\phi$ equation becomes, for $X = e^{\Phi-\Lambda}h_1/r$ and $dr_* = e^{\Lambda-\Phi}dr$,

$$\frac{d^2 X}{dr_*^2} + [\omega^2 - V] X = 0, \quad V = e^{2\Phi} \left[\frac{\ell(\ell+1)}{r^2} - \frac{6m}{r^3} + 4\pi(\rho - p_r) \right], \quad (1)$$

after using $m' = 4\pi r^2 \rho$, $\Phi' = (m + 4\pi r^3 p_r)/r(r - 2m)$ and the anisotropic TOV equation (sympy: difference 0). The tangential pressure and the fluid velocity drop out, which is the thick-matter analogue of the first round's statement that the odd junction is independent of the equation of state. For a shell ($p_r = 0$), the anisotropic TOV equation gives $p_t = \rho r \Phi'/2$, and integrating over the layer gives, as the thickness tends to zero, surface density $\int \rho d\ell = (1 - \sqrt{f_R})/4\pi R = \sigma$ and surface pressure $\int p_t d\ell = (1 - \sqrt{f_R})^2/(16\pi R \sqrt{f_R}) = p$: exactly model C. The potential contains $\int 4\pi \rho e^{2\Phi} dr_* = \sqrt{f_R} 4\pi\sigma$, i.e. a δ -function of strength $\Delta_{\text{odd}} = 4\pi\sigma\sqrt{f_R}$, while the other terms stay bounded. Since $X = (\omega/2i)\Psi_{\text{CPM}}$ on both sides (with the interior normalised to its own time), this reproduces Eqs. (4.8)–(4.9) exactly.

6.3 Numerical thick-shell limit

In `c03b` a smooth shell of half-thickness ε ($\rho \propto [1 - (r - R)^2/\varepsilon^2]^2$) was built numerically, X was integrated from the flat interior through the matter, and the direction of (X, X') at $r = R + \varepsilon$ was compared with the thin-shell prediction built from the first round's matrix $\mathbf{T}_{\text{odd}}$ and from three wrong alternatives. Fig. 2: only the first round's matrix gives a mismatch that tends to zero, linearly in ε (6.6×10^{-4} at $\varepsilon = 0.006M$, $\omega M = 0.2$), while the alternatives saturate at $4\text{--}10 \times 10^{-2}$ (O3 **CONFIRMED**).

6.4 Literature

PSP26 Eq. (5.23), $F^{1/2}\eta_{\text{out}} - \eta_{\text{in}} + F^{1/2} - 1 = 0$, is on p. 20 of the PDF (we read it). With $\eta = R\hat{\psi}'/\hat{\psi}$ of the retarded-time functions $\hat{\psi}_{\text{out}} = e^{-i\omega r_*}\Psi$ and $\hat{\psi}_{\text{in}} = e^{-i\omega r}\Psi$, the extra terms are $F^{1/2}(-i\omega R/F) - (-i\omega R) = 0$ because $w = F^{-1/2}\omega$; (5.23) is then identical to (4.8). Pani et al. (3.29) and (A35) are $[[h_0]] = 0$ and $[[\sqrt{h}h_1]] = 0$ (read in the arXiv v2 PDF), and their text states that for continuous h these imply continuity of Ψ , Ψ' (O4 **CONFIRMED**).

7 Even-parity junction, model C

The even derivation (a symbolic brute-force computation of ~ 2.5 MB of expressions) was not repeated independently. It was tested through its consequences, which is sufficient to validate it within the precision of the tests.

- **Determinant, reality, flux** (c06). Over 300 random (ω, ℓ, R, v_s^2) with $R \in [2.35, 30]M$: $\max |\det T\sqrt{f_R} - 1| = 1.1 \times 10^{-12}$ (real ω) and 9.2×10^{-13} (complex ω); $\text{Im } T = 0$ exactly for real ω ; the flux $J = \text{Im}(\Psi^* \partial_x \Psi)$ is conserved to 1.6×10^{-13} for arbitrary interior data (E1, E2 **CONFIRMED**).
- **Transparent limit**: $|T^C - \mathbb{I}| = 4.0 \times 10^{-5}, 4.0 \times 10^{-7}, 4.0 \times 10^{-9}$ at $M = 10^{-4}, 10^{-6}, 10^{-8}$ (E3 **CONFIRMED**).
- **Precision near $2M$** (E4 **PARTLY**): in double precision $|\det T\sqrt{f_R} - 1| = 10^{-9}$ ($2.3M$), 1.3×10^{-7} ($2.2M$, the text says 10^{-6}), 1.6×10^{-5} ($2.1M$), 5×10^{-3} ($2.05M$), 0.87 ($2.02M$); the 40-digit path gives $\leq 3 \times 10^{-15}$ down to $2.01M$, as claimed.
- **Poles** (E5 **PARTLY**): the zeros of $\det A(\omega)$ (poles of T) lie at $\omega(R^3/M)^{1/2} \approx 1.54$ and $0.985i$ ($R = 6M$) – at the matter scale, but not at the matter QNM frequencies ($1.315, 0.590i$); “poles at the frequencies of the matter modes” is loose.
- **PSP26 matter modes** (E6 **CONFIRMED**). Reading PSP26 Figs. 1–2 (p. 3–4) we find, for $M/R = 0.3 \dots 0.01$, $\text{Im } \omega(R^3/M)^{1/2} \approx 0.668, 0.637, 0.608, 0.582, 0.557, 0.535, 0.518$ and $\text{Re } \approx 1.096, 1.188, 1.267, 1.336, 1.398, 1.454, 1.495$, $\text{Im } \approx -1.70, -1.51, -1.10, -0.65, -0.28 (\times 10^{-3})$: the first round’s table agrees at the reading precision (± 0.002). Our independent solver reproduces every entry of that table to $\leq 2 \times 10^{-11}$. Beyond the first round, we compared the Γ dependence at $M/R = 0.2$ (PSP26 Figs. 1–2, right/lower panels): at $\Gamma = 1.81$ and 2.46 we obtain $0.6437i, 1.1383 - 8.2 \times 10^{-4}i$ and $0.5511i, 1.5486 - 1.93 \times 10^{-3}i$, against the plotted $0.643, 1.14, -0.82 \times 10^{-3}$ and $0.551, 1.548, -1.93 \times 10^{-3}$.
- **Newtonian limit, independent derivation** (c05). We solved from scratch the Newtonian problem of a self-gravitating 2D fluid shell (continuity, Laplace-type normal force $2Hp$, self-gravity including the multipole shift of the displaced mass sheet, $\delta p = \Gamma p \delta \sigma / \sigma$). The 2×2 eigenproblem for $\zeta^2 = \omega^2 R^3 / M$ has trace $[(\ell^2 + \ell + 4)\Gamma - (\ell^2 + \ell + 6)]/4$ and determinant $-(\ell - 1)\ell(\ell + 1)[(\ell + 2)(2\ell - 3)\Gamma - 4(\ell - 2)]/[16(2\ell + 1)]$; its roots coincide identically (sympy) with PSP26 Eq. (7.8) for all ℓ, Γ . For $\ell = 2$, $\Gamma = 2$: $\zeta = 1.504962$ and $0.514695i$. The first round’s full-GR values at $M/R = 0.01, 0.02$ extrapolate (Richardson) to $0.51463i$ and 1.50514 , i.e. to this limit.
- **Misprint in PSP26**: the quartic (7.6) printed in PSP26 has constant term $-(\ell - 1)\ell(\ell + 1)(\ell + 2)[(2\ell - 3)\Gamma - 4]/[16(2\ell + 1)]$, which is inconsistent with its own solution (7.8) and with its Newtonian Eq. (10.62); for $\ell = 2, \Gamma = 2$ it would give no unstable mode. The correct constant is the determinant above. The first round used (7.8) and is unaffected.
- **Wave modes** (E7 **CONFIRMED**): the five entries of `psp_wave.tex` agree with our reading of PSP26 Figs. 3–4 (e.g. odd, $M/R = 1/6$, mode 2: $\text{Re}(R\omega) \approx 4.15$, $\text{Im}(R\omega)/\xi \approx -0.83$). PSP26’s Newtonian formulas (7.23)–(7.25) are quoted correctly.
- **High frequency** (E8 **CONFIRMED**). Although T_{21}^C does not tend to the odd value (the ratio of the implied δ strengths is $0.455, 0.890, 1.105$ at $R = 3, 6, 10M$), the $O(1/\omega)$ off-diagonal

Even-parity matter modes of the static fluid shell (model C, $\ell = 2$)

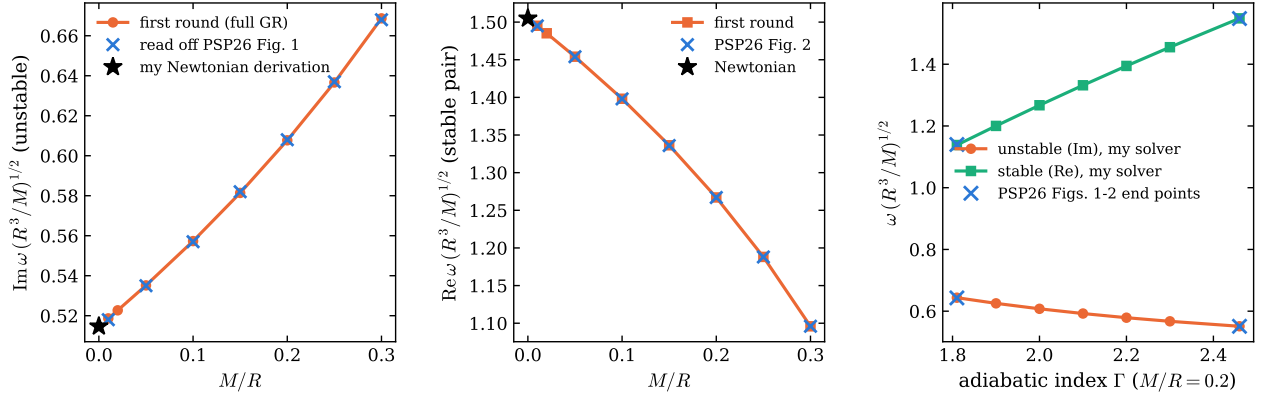


Figure 3: Matter modes of the static fluid shell: first round, values read off PSP26, our Newtonian limit and our Γ scan.

terms compensate, and the reflectivities do merge: $\mathcal{R}_{\text{even}}/\mathcal{R}_{\text{odd}} = 1.0097, 1.0002, 1.0000$ at $\omega M = 2, 5, 10$ ($R = 3M$).

8 Model A: the frozen domain wall

What is confirmed. The overdetermination (8 equations, 6 unknowns, rank 6 vs 8) is in the first-round log. With the first round’s prescription (all four metric-continuity rows plus the trace and trace-free angular Israel rows) the relative constraint violation falls as ω^{-1} and $|\mathcal{R} + \mathcal{T} - 1|$ as ω^{-2} (local exponents -2.00 between $\omega M = 2$ and 32; A2 **CONFIRMED**). The kinematic statement $\ddot{T}_0 \propto \dot{R}^2$ follows from $\dot{T}_0 = \beta/f$, $\partial \dot{T}_0 / \partial \dot{R} = \dot{R}/f\beta$ (A3 **CONFIRMED**).

What fails (A4, A5 **INCORRECT).** The first round’s justification of its prescription is that the $\tau\tau$ and τA rows are “constraints” which an exact solution would satisfy automatically. For the frozen ansatz, which is not a solution, every choice of two Israel rows is equally an approximation. We kept the metric-continuity rows (the first junction condition must hold) and tried all six choices (c06d, c06e, Fig. 4):

imposed Israel rows	$\mathcal{R}(\omega M = 30), R = 3M$	$ \mathcal{R} + \mathcal{T} - 1 $	residual of dropped rows	lowest QNM, $R = 6M$
trace + trace-free (first round)	6.8×10^{-5}	1.7×10^{-4}	1.9×10^{-4}	$0.18120 - 0.22393i$
$\tau\tau$ + trace-free	1.8×10^{-6}	10^{-13}	0.26	$0.13978 - 0.27050i$
τA + trace-free	3.4×10^{-6}	5.6×10^{-5}	0.13	$0.17063 - 0.26403i$
$\tau\tau$ + trace	1.000	3.9×10^{-9}	2.1×10^{-5}	(none near)
τA + trace	0.9998	2.8×10^{-8}	2.1×10^{-5}	(not computed)
$\tau\tau$ + τA	0.9999	10^{-13}	6.7×10^{-5}	$0.16818 - 0.19045i$

(For comparison: odd parity and model C give 1.8×10^{-6} and fundamental modes $0.24739 - 0.22534i$ and $0.13710 - 0.26885i$.) The predictions differ at $O(1)$, and the differences do not shrink with ω . Three observations settle the matter: (i) with the first round’s choice the energy error exceeds $\mathcal{R}$ itself at every high frequency (3.8×10^{-2} vs 1.6×10^{-2} at $\omega M = 2$; 1.5×10^{-4} vs 6.0×10^{-5} at $\omega M = 32$, $R = 3M$): the quoted $\mathcal{R}$ of model A has no significant digit; (ii) choices that satisfy the dropped rows better (e.g. $\tau\tau$ +trace, residual 2×10^{-5}) predict total reflection instead; (iii) the choice $\tau\tau$ +trace-free conserves energy exactly and reproduces the universal high-frequency law $\mathcal{R} \simeq \Delta_{\text{odd}}^2/4\omega^2$ shared by odd parity and model C. Consequently the statements “Model A ... reflects much more strongly, and its $\mathcal{R}$ decays more slowly” (derivation Sec. 6.2 item 5, Figs. 2, 11) and “its even-parity results should be read as $O(\omega_{\text{dyn}}/\omega)$ -accurate” are not supported. Together with Sec. 4 (B12) – the wall’s QNMs have $|\omega| \approx 2\text{--}5 \omega_{\text{dyn}}$ – the even-parity model-A spectrum and reflectivity should be regarded

Model A is prescription dependent: which 2 of the 4 Israel equations are imposed (metric continuity always imposed)

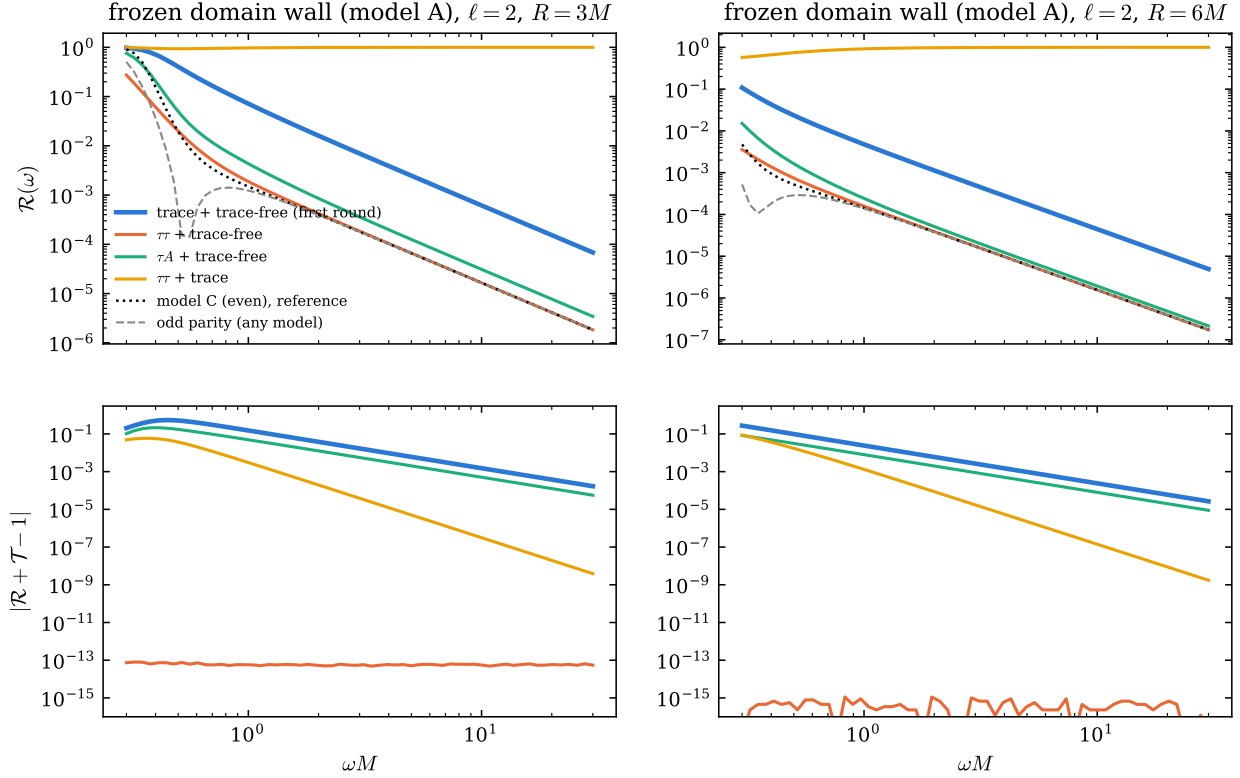


Figure 4: Reflectivity and energy balance of the frozen domain wall for four admissible prescriptions (metric continuity always imposed). The first-round prescription (thick blue) is not singled out by any criterion.

as undetermined by the frozen approximation. The odd sector of model A is unaffected (exactly consistent, O5).

9 Numerical methods and results

Exterior machinery. The continued-fraction coefficients reproduce Pani et al. (B4)/(B7) under $\omega \rightarrow -\omega$ (their time dependence is $e^{+i\omega t}$; we compared the printed c_i, d_i, e_i). Their convergence condition ($R_2/2 < a < R_2$, “ $R_2 > 2$ ” in units $2M = 1$) is equivalent to the first round’s $R_2 \geq 4M$ (N1 **CONFIRMED**). The asymptotic recursion (5.5) was re-derived symbolically for $k = 1 \dots 5$ (N2 **CONFIRMED**).

Scattering coefficients (c04). Our solver recomputed $\mathcal{R}$ and $\mathcal{T}$ at 12 frequencies of nine configurations (odd $\ell = 2, 3$; even-C $\ell = 2, 4$; even-A $\ell = 2$; $R = 2.2\text{--}6M$). Largest differences from the first round’s Method 1: 1.5×10^{-11} in $\mathcal{R}$ and 1.3×10^{-8} in $\mathcal{T}$, the latter set by our own $|\mathcal{R} + \mathcal{T} - 1| \sim 5 \times 10^{-9}$. The first round’s table gives $\max |\Delta_{12}| = 6 \times 10^{-10}$; the text’s “ $\lesssim 10^{-10}$ ” should read “ $\lesssim 10^{-9}$ ” (N3 **PARTLY**).

QNMs (c04). 67 modes (51 catalogue modes – wave and matter modes of odd, even-C, even-A for $\ell = 2, 3$ and $R = 2.2\text{--}10M$ – and the 16 matter modes of the PSP table) were re-converged with our solver from the first-round values: all wave and growing modes agree to $10^{-10}\text{--}10^{-13}$ (1.6×10^{-7} for the strongly damped $0.78377 - 0.71437i$ at $R = 2.2M$); the damped imaginary matter modes agree to $10^{-5}\text{--}10^{-7}$, the same level as the first round’s own Method-1/2 difference (N5 **CONFIRMED**).

Of the 397 catalogue entries, 28 have $|w_1 - w_2| > 10^{-8}$ and 8 exceed 10^{-6} (max 6.4×10^{-5}); all are below the task's 10^{-4} , but the claims of 10^{-8} – 10^{-13} are overstated (N4).

Completeness of the catalogue (N6 **PARTLY**).

- $\ell = 2$, even-C: the stable matter pair is absent at $R = 2.2, 2.5, 3M$, although it exists and is continuous with the tracked mode: $0.185919 - 0.000030i$ ($2.2M$), $0.212774 - 0.000206i$ ($2.5M$), $0.197121 - 0.000310i$ ($3M$). At $R = 3M$ the entry labelled “matter-stable” is a duplicate of the unstable mode ($+0.13321i$).
- $\ell = 4$, $R = 2.2M$: the unstable matter mode ($0.43726i$) is missing; the entry labelled “matter-imag-damped” ($1.27806 - 0.72384i$) is a wave mode.

Tables `shell_qnm_12/14.tex` and `qnm_table.csv` inherit these omissions. They do not affect the physical conclusions (the instability is present for all ℓ and R examined).

Other items. The Boyanov et al. estimator is implemented as $|i\omega + \Psi_{,r_*}/\Psi|/|i\omega - \Psi_{,r_*}/\Psi|$, whose square is compared with $\mathcal{R}$; the agreement $2\text{--}4 \times 10^{-6}$ is consistent with the neglected $O(1/r)$ terms (N7 **CONFIRMED**). In Fig. 5 the estimator's crosses are hidden under the circles. The Method-1 NaN at $\ell = 4$, $\omega M = 0.00566$ is present for all three $R = 6M$ curves, as stated (N8 **CONFIRMED**).

10 Schwarzschild validation, WKB and limits

QNMs (c02). Our 30-digit Leaver solver with n -th inversion gives $\omega M = 0.3736716844 - 0.0889623157i$ for $\ell = 2$, $n = 0$ and agrees with the first round's twelve values to $\leq 1.4 \times 10^{-16}$, and with the six-digit table of Berti, Cardoso & Starinets to $\leq 6 \times 10^{-7}$ (rounding) (S1, S2 **CONFIRMED**).

WKB (S3, S4). The third-order WKB was recomputed from the explicit Iyer–Will formula (1.5a,b) (coded independently from the hypervirial construction): all twelve values agree with the first round to 5×10^{-16} , which validates the hypervirial code at that order; $\ell = 2$, $n = 0, 1$ give $0.37316 - 0.08922i$ and $0.34602 - 0.27492i$, consistent with the values quoted from Iyer–Will. The sixth-order values could not be compared with Konoplya's table (full text not accessible); they behave as expected (error 1.4×10^{-7} for $\ell = 4$, $n = 0$) (**NOT VERIFIABLE**). The errors quoted in the text for $\ell = 2$, $n = 0$ (7×10^{-2} , 1.5×10^{-3} , 2.3×10^{-4}) are relative errors $|\Delta\omega|/|\omega|$; the absolute errors are 2.5×10^{-2} , 5.7×10^{-4} , 8.9×10^{-5} .

Black-hole reflectivity (S5 **CONFIRMED**). With an integrator in r_* (LSODA, start at $r_* = -80M$) we obtain $\mathcal{R} = 1.548455 \times 10^{-2}$ at $(2M\omega)^2 = 1$ and 5.3117×10^{-9} at $(2M\omega)^2 = 4$, equal to the first round's values; the scanned Vishveshwara Fig. 2 is consistent (about 1.5×10^{-2} and $\sim 10^{-8}$).

Limits (c08). Low frequency: $1 - \mathcal{R} \leq 10^{-15}$ at $\omega M = 10^{-3}$ for odd, even-C, even-A (L1 **CONFIRMED**). High frequency: $\mathcal{R}/[\Delta^2/(4\omega^2 + \Delta^2)] = 0.997\text{--}0.999$ at $\omega M = 8$ (L2 **CONFIRMED**). Closed system: $||S| - 1| \leq 4 \times 10^{-16}$ (L4 **CONFIRMED**). Black-hole limit (L3 **PARTLY**): the approach is real but slow and oscillatory, the amplitude of the shell contribution scaling like $\Delta_{\text{odd}} \propto (R - 2M)^{1/2}$; at $\omega M = 0.6$, $\mathcal{R}/\mathcal{R}_{\text{BH}} = 8.4, 4.8, 3.1, 0.32, 0.89$ for $R - 2M = 0.2, 0.05, 0.01, 0.002, 0.0002$ (Fig. 5), so the table value at $2.002M$ (0.00021 vs 0.00066) is not “consistent” in the sense of convergence yet. Trapped modes: all seven values of `trapped.tex` reproduced to $\leq 4 \times 10^{-12}$; WKB real parts are off by 0.6–6.3% and imaginary parts by factors 1.12–2.03 (the text says 1–6% and $\lesssim 2$) (L5 **PARTLY**).

11 Literature and citations

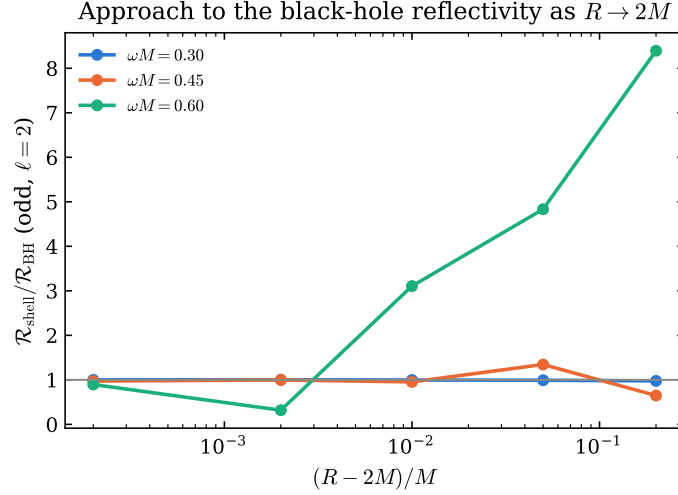


Figure 5: Oscillatory approach of the odd-parity shell reflectivity to the black-hole value.

Table 3: Every reference of the first round.

reference	what the first round uses	status
Regge–Wheeler 1957	Eq. (25) (k_{eff}^2 , i.e. the odd potential), (24), gauge (18), (20)	local PDF (page read); CONFIRMED
Zerilli 1970	Eq. (5), first-order system, Eq. (3)	local PDF; CONFIRMED
Vishveshwara 1970	Fig. 2	local PDF (scan); CONFIRMED
Martel–Poisson 2005	Eqs. (3.1)–(3.7), (4.1)–(4.3), (4.23)–(4.26), (5.1)–(5.18)	local PDF, all numbers exist; CONFIRMED
Martel thesis	“consulted”	local PDF; not used
Ipser–Sikivie 1984	(2.1)–(2.14), (3.1)–(3.9), quote	local PDF (pages read); CONFIRMED
Poisson 2004	(3.49), (3.56), (3.65), (3.68)–(3.70), (3.75)–(3.81)	local PDF, all numbers exist; CONFIRMED
Abramowitz–Stegun; DLMF 10.50.1	Riccati–Bessel functions; Wronskian	identity verified numerically; CONFIRMED
Pani et al. 2009	(3.29), (3.33), (A16), (A20), (A35), App. B	arXiv v2 PDF; labels and content CONFIRMED . Full title has “I. Nonradial oscillations . . .”
Pitre–Schneider–Poisson 2026	(1.1)–(1.2), (2.4)–(2.6), (5.23), (7.8), (7.23)–(7.25), Figs. 1–4	arXiv PDF; PRD 113, 124055 (2026). CONFIRMED ; $\delta p = \Gamma p \delta \mu / (\mu + p)$ is PSP26 (2.7)
Boyanov et al.	App. B1, Eqs. (14), (15), (17), Fig. 1	arXiv; PRL 135, 151402 (2025) (INSPIRE). CONFIRMED
Leaver 1985	continued fraction	INSPIRE; CONFIRMED
Berti–Cardoso–Starinets 2009	QNM tables	INSPIRE; values CONFIRMED

Table 3: Every reference of the first round.

reference	what the first round uses	status
Leins–Nollert–Soffel 1993	continued fraction about R_2	cited as such by Pani et al.; CONFIRMED
Schutz–Will 1985;	WKB	INSPIRE; order 3 verified;
Iyer–Will 1987;		Iyer’s companion paper II
Konoplya 2003		(PRD 35, 3632) contains the Schwarzschild tables
Hatsuda 2020;	WKB = Bender–Wu; hypervirial	ADS/APS; CONFIRMED
Killingbeck 1978		
Chandrasekhar 1983	Sec. 4.26, $RW \leftrightarrow Z$ map	book not available; map verified; NOT VERIFI-ABLE (section number)
Cardoso–Franzin–Pani 2016	echoes of ultracompact objects	well known PRL 116, 171101; CONFIRMED
Yang, Bonga, Pen	eikonal instability	INCORRECT : H. Yang, B. Bonga, Z. Pan, PRL 130, 011402 (2023)

The honesty statement (PSP26 computed the QNMs of the same system; no published $\mathcal{R}$, $\mathcal{T}$ found) is consistent with our own reading of PSP26 (no reflection coefficients in the parts we read) and a web search.

12 Compliance with the task statement

All required products exist: derivation with the six requested sections; potentials, $\mathcal{R}/\mathcal{T}$, method comparison, complex-plane QNMs, QNMs vs R/M , energy-conservation plots; a CSV with the requested columns and a literature flag. DOP853 with $\text{rtol}=10^{-10}$, $\text{atol}=10^{-12}$ is used. The requirement that *every* function carry a docstring citing its equation is only partly met: of 144 functions (including nested helpers) 57 have no docstring and 59 more cite no source (C5 **PARTLY**); the main physics routines are documented. The first round’s corrections to the task statement (Option B does not exist; Option C has no free κ ; interior frequency ν ; the $\sigma \rightarrow 0$ limit is flat space and the black-hole reflectivity is reached as $R \rightarrow 2M$ instead) are all correct.

13 Integrity of the first round

The SHA-256 hashes of all 102 files of `First round` (Python caches excluded) were recorded before the verification (`first_round_sha256_before.txt`) and recomputed at the end (2026-09-25 15:35, `first_round_sha256_after.txt`). The two lists are **identical**. No file of the first round was created, modified or deleted by the second round; all executions used copies inside `Second round`.

14 Recommendations

1. Remove or heavily qualify the even-parity model-A results (reflectivity, QNMs, Figs. 2, 6, 11 and the statements about the wall reflecting more strongly); if kept, show the prescription dependence of Fig. 4.
2. Correct the time-scale statements of the Discussion (the wall collapses on $\sim R/\sqrt{2}$).
3. Replace “ $\lesssim 10^{-8}$ ” / “ 10^{-8} – 10^{-13} ” by the actual method agreement ($\leq 6.4 \times 10^{-5}$ overall, $\lesssim 10^{-10}$ for wave modes); state that WKB errors are relative.

4. Complete the catalogue (stable matter pairs at $R \leq 3M$, $\ell = 4$ unstable mode at $2.2M$) and fix the labels.
5. Rephrase the Option-B argument (Sec. 4); correct “U.-L. Pen” to “Z. Pan”; mention the misprint in PSP26 (7.6).